

Monetary Policy in Fractured Markets:

A Unified HANK-IO Framework with Structural

Monopsony

and Geoeconomic Fragmentation

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Abstract

We build a quantitative HANK model with structural monopsony and multi-bloc input-output networks, solved via sequence-space Jacobians. Three results emerge. First, countercyclical wage markdowns endogenously flatten the Phillips curve (slope falls 1.4% in baseline). Second, geoeconomic fragmentation concentrates Domar weights in high-monopsony upstream sectors, amplifying wage suppression non-linearly; a 15pp fragmentation shock reduces TFP by 0.47%. Third, the welfare-optimal Taylor rule accommodates supply-side inflation ($\phi_\phi^* = -0.22$), not fights it. A commuting-zone Bartik design confirms the IO $\times$ Monopsony wage interaction in 2021–2023 microdata.

JEL: E12, E24, E52, F41, J42.

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1. Introduction

Consider two commuting zones in 2022: a rural Appalachian mining county and a suburban professional services corridor. Both face the same Federal Reserve rate hikes. But only the mining county—heavily employed in upstream sectors dependent on cross-bloc intermediate inputs, with a labor market HHI above 4,000—sees real wages fall by 3% over 2022–2023. The services corridor sees wages fall by less than 1%. Standard monetary models predict the same rate-hike transmission for both. This paper builds and estimates a model that predicts the difference.

The core mechanism is a non-linear interaction that requires three simultaneous ingredients: a multi-region input-output network (the mining county’s upstream exposure to fragmented supply chains), structural monopsony (the concentrated labor market that amplifies wage compression), and household heterogeneity (the hand-to-mouth workers in the mining county who cannot smooth the income loss). Remove any one ingredient, and the differential wage response disappears. We document this interaction in commuting-zone microdata, then build the framework needed to understand its aggregate and distributional consequences.

The Monopsony Fact. [Azar et al. \(2022\)](#), [Rinz \(2022\)](#), and [Benmelech et al. \(2022\)](#) document that the median U.S. worker faces a local labor market Herfindahl-Hirschman Index (HHI) exceeding 4,000—the Department of Justice’s threshold for “highly concentrated.” Wage markdowns (the gap between the marginal product of labor and the observed wage) average 15–30% across sectors, with substantial cross-sectional variation positively correlated with HHI ([Azar et al., 2022](#)). Critically, this gap is *countercyclical*: firms compress wages more aggressively in downturns, when workers’ outside options deteriorate and firm-level labor supply elasticities fall ([Dube et al., 2020](#); [Manning, 2021](#)). This decouples unemployment from inflation in ways standard models cannot reproduce, and it implies that the aggregate Phillips curve is *endogenously flatter* than competition-based calibrations suggest.

The Fragmentation Fact. The IMF’s Geoeconomic Fragmentation series ([Aiyar et al., 2023](#)) documents a measurable bifurcation of global trade into politically aligned blocs, accel-

erating sharply post-2022. Using bilateral trade flow data, [Gopinath et al. \(2024\)](#) show that trade within Western-aligned and China-aligned blocs has remained resilient, while cross-bloc trade has been rerouted through costly intermediary countries or reshored domestically. This structural shift means supply shocks that were previously absorbed by flexible global substitution are now amplified domestically and propagate through concentrated upstream networks with elevated Domar weights ([Baqae and Farhi, 2019](#)).

The Critical Non-Linearity. These two facts interact in a way that *cannot* be captured by studying them in isolation—and this interaction is the central contribution of the paper. Trade fragmentation shifts production toward domestic upstream sectors: Energy, Mining, and Basic Manufacturing absorb the largest Domar weight increases when cross-bloc trade costs rise. These same sectors are precisely where monopsony is most severe. Firm-level labor supply elasticities in goods production average $\varepsilon \approx 2.5$, less than half the service-sector average of 5.5 ([Dube et al., 2020](#)). Because the wage markdown deepens as $1/(\varepsilon + 1)^2$, a fragmentation shock that reallocates activity toward high-monopsony sectors triggers wage compression that is approximately $[(5.5 + 1)^2 / (2.5 + 1)^2] \approx 3.4$ times larger per unit Domar weight reallocation in goods than in services. In our calibration, a 25pp tariff shock reduces real wages in goods sectors by 3.2% versus 1.4% in services. An IO model without monopsony misses this dispersion entirely (it predicts proportional wage effects). A monopsony model without IO misses the concentration of Domar weight in high-monopsony sectors and therefore understates the aggregate wage compression by a factor of two. Only the joint model correctly identifies the distributional incidence of fragmentation shocks and, as a consequence, the correct augmented Taylor rule response.

The Central Gap. No existing general equilibrium model integrates both of these facts in a dynamic, quantitative framework. HANK-IO models ([Auclert et al., 2021b](#); [Baqae and Farhi, 2019](#)) handle heterogeneous agents and input-output networks but assume competitive labor markets and integrated trade. Monopsony has been grafted onto static or representative-agent NK frameworks ([Berger et al., 2022](#); [Autor et al., 2020](#)), but without the heterogeneous agent redistribution channel or the multi-region network structure. This

paper fills that gap.

1.1. What We Do

We build a model with three novel, interacting components:

1. **Structural monopsony with a countercyclical markdown.** Firms face upward-sloping firm-level labor supply curves, parameterized by a firm-level elasticity ε that falls when unemployment rises. The resulting wage markdown $\mu^w = \varepsilon/(\varepsilon + 1)$ deepens in recessions, generating endogenous, state-dependent Phillips curve flattening.
2. **Multi-region, multi-bloc IO network.** The world is partitioned into geopolitical blocs. Trade costs between blocs are higher and subject to a time-varying *fragmentation premium* ϕ_t that follows an AR(1) process with high persistence. Firms source intermediates using a nested CES structure over domestic and foreign varieties; the Armington elasticity η governs substitution across regions. We represent the resulting multi-sector production structure as a weighted directed graph and compute Domar weights and Leontief inverses analytically.
3. **Heterogeneous households with incomplete markets (HANK).** A continuum of households differs in asset positions and labor types. They face idiosyncratic income risk, borrow against a constraint, and consume out of current income with heterogeneous marginal propensities to consume (MPCs). The distribution of assets and labor types is a state variable.

We solve the model using the *sequence-space Jacobian* (SSJ) method of [Auclert et al. \(2021a\)](#), extended with two novel Jacobian blocks: (i) the *monopsony block*, which adds the countercyclical markdown response to the standard wage-employment Jacobian; and (ii) the *fragmentation block*, which maps shocks to the trade cost matrix $\tau(\phi_t)$ into shocks to the full $(S \times R)$ -dimensional Leontief inverse. Both extensions are analytically tractable and preserve the sparse structure required for efficient computation.

1.2. Main Results

Proposition 1 (Phillips Curve Flattening). The monopsony-adjusted NKPC slope is:

$$\kappa^{mono} = \kappa_{mc} \left(\frac{1}{\sigma} + \frac{\varphi}{\alpha_n} - \frac{\gamma}{(\bar{\varepsilon} + 1)^2 (1 - \bar{u}) \alpha_n} \right) < \kappa^{comp},$$

where $\kappa_{mc} = (1 - \xi)(1 - \beta\xi)/\xi$ is the Calvo coefficient on real marginal cost. The second term in the bracket—absent in the competitive model—reflects the countercyclical markdown channel: when employment rises, firm-level labour supply becomes more elastic (ε_t rises), the markdown falls, and the wage rises by less than the marginal product, damping marginal cost pass-through to prices. In our baseline calibration ($\bar{\varepsilon} = 4$, $\gamma = 0.8$, $\alpha_n = 0.65$), the flattening is 1.4% ($\kappa^{mono} = 0.307$, $\kappa^{comp} = 0.311$). The mechanism is qualitatively robust across parameterisations; quantitative magnitudes scale with γ and rise substantially for lower values of $\bar{\varepsilon}$ (higher monopsony concentration).

Proposition 2 (Fragmentation Amplification). The semi-elasticity of aggregate TFP with respect to the fragmentation premium is:

$$\frac{d \log \text{TFP}}{d\phi} = - \sum_{\substack{(s,r),(s',r') \\ b(r) \neq b(r')}} \lambda_{sr} \Omega_{(s,r)(s',r')} \frac{1}{\tau^{rr'}},$$

a closed-form, network-weighted expression. This quantity is strictly negative and increasing in magnitude in the centrality of cross-bloc linkages. In our calibrated 10-sector, 5-region model with the 2019 BEA Use Table, a 15pp increase in ϕ reduces aggregate TFP by 0.47%, producing a stagflationary impulse—output falls while marginal costs rise—that standard Taylor rules address inefficiently by raising rates and further depressing output.

Proposition 3 (Optimal Augmented Taylor Rule). The welfare-optimal monetary rule is:

$$i_t = r^* + \pi^* + \phi_\pi^* (\pi_t - \pi^*) + \phi_y^* \tilde{y}_t + \underbrace{\phi_\mu^* \hat{\mu}_t^w}_{\text{novel: } >0} + \underbrace{\phi_\phi^* \hat{\phi}_t}_{\text{novel: } <0} + \nu_t.$$

The sign of $\phi_\mu^* \geq 0$: monopsony deepening compresses high-MPC workers’ incomes below competitive levels; the central bank can partially offset this through accommodation. The sign of $\phi_\phi^* < 0$: fragmentation spikes are supply shocks—standard Taylor rules tighten into the output loss; the augmented rule ($\phi_\phi^* = -0.22$) trades a modest inflation overshoot for substantially less output sacrifice.

Relation to the literature. We build on three bodies of work. On heterogeneous agents: [Auclert \(2019\)](#) and [Kaplan et al. \(2018\)](#) show redistribution channels dominate monetary transmission in HANK; we add the countercyclical monopsony channel as a novel source of MPC amplification. On IO networks: [Baqaee and Farhi \(2019\)](#) and [Acemoglu et al. \(2012\)](#) price sectoral shocks via Domar weights; we extend this to a multi-bloc setting where the Leontief inverse varies with geopolitical fragmentation. On labor market power: [Berger et al. \(2022\)](#) introduce monopsony into a static NK model; we are the first to embed countercyclical monopsony in a dynamic HANK-IO framework and derive its implications for optimal monetary policy.

1.3. Organization

Section 2 establishes the CZ-level Bartik evidence. Section 3 presents the model. Section 4 defines equilibrium and key log-linearizations. Section 5 describes the SSJ method. Section 6 proves the three propositions. Section 7 presents quantitative exercises. Section 8 draws policy implications. Section 9 concludes. A Supplemental Appendix contains computational details, robustness checks, and extensions.

2. Motivating Evidence: Fragmentation, Concentration, and Wage Suppression

We begin by establishing the empirical fact that motivates the model: geoeconomic fragmentation suppresses wages in local labor markets, and this suppression is disproportionately concentrated in markets with high monopsony power. This non-linear interaction—which neither fragmentation alone nor monopsony alone can generate—is the central stylized fact the model is designed to explain.

2.1. Data

Wage and employment data. We use the *Quarterly Census of Employment and Wages* (QCEW), which provides establishment-level employment and wage bills by NAICS industry and county from 1990–2023. We aggregate to commuting zones (CZs) following the [Autor et al. \(2013\)](#) crosswalk (741 CZs covering the contiguous US), using the NAICS–BEA sector mapping from the Supplemental Appendix to merge with the IO network. The wage outcome is $\Delta \ln \bar{w}_{c,i}$: the log change in average weekly wages in CZ c , industry i , between the pre-period (2018–2020 average) and post-period (2022Q2–2023Q4, the Fed hiking cycle coinciding with peak fragmentation exposure).

Labor market concentration. Following [Azar et al. \(2020\)](#) and [Azar et al. \(2022\)](#), we measure local monopsony using the occupation-by-CZ Herfindahl-Hirschman Index (HHI) constructed from online job postings (Lightcast/Burning Glass Technologies, 2017–2020 pre-period average). For CZ c and sector i , $\text{HHI}_{c,i}$ is the employment-weighted average posting HHI across occupations within that sector-CZ cell. The pre-period average is used to ensure exogeneity of the concentration measure with respect to post-2021 shocks.

IO network and upstream exposure. We use the BEA 2019 Benchmark Use Table aggregated to our 10-sector classification (Supplemental Appendix). For each sector i , define its *cross-bloc upstream exposure* as the total cost share of inputs sourced from sectors with significant Eastern-bloc supply links:

$$E_i^{\text{upstream}} = \sum_{j \in \mathcal{S}^{\text{cross}}} \Omega_{ij}, \quad (1)$$

where $\mathcal{S}^{\text{cross}} = \{\text{Energy, Mining, Technology Hardware, Chemicals \& Pharma}\}$ are the sectors with the highest cross-bloc Domar weights (verified from the calibrated IO model in Section 3.3).

Broadened fragmentation index. Standard measures capture tariff and iceberg trade costs. Section 2.5 below argues that a complete measure of geoeconomic fragmentation must

also incorporate IP and regulatory decoupling—export controls on semiconductor technology (BIS Entity List, October 2022 chip controls), cross-border patent licensing disruptions, and investment screening (CFIUS actions, EU FDI screening regulation). We construct a *composite fragmentation index* F_t that combines:

- (i) The IMF Geoeconomic Fragmentation index ([Aiyar et al., 2023](#)) (trade flows);
- (ii) The PIIE Trade Policy Uncertainty index ([Gopinath et al., 2024](#)) (tariff and non-tariff barriers);
- (iii) The count of BIS Entity List additions per quarter (regulatory fragmentation), normalized to unit variance.

The composite F_t has quarterly correlation 0.87 with the pure trade-cost measure but loads significantly more on the 2022Q4 chip export control shock.

2.2. Local Fragmentation Exposure: A Bartik Design

We construct a local fragmentation exposure index using the standard shift-share (Bartik) framework ([Goldsmith-Pinkham et al., 2020](#)). For each CZ c :

$$\text{Exposure}_c = \sum_{i=1}^{10} \underbrace{s_{c,i}^{\text{pre}}}_{\text{local share}} \times \underbrace{E_i^{\text{upstream}}}_{\text{national shift}} \quad (2)$$

where $s_{c,i}^{\text{pre}} = L_{c,i,\text{pre}}/L_{c,\text{pre}}$ is the CZ’s pre-2021 employment share in sector i , and E_i^{upstream} is the national upstream cross-bloc exposure (1). The identifying variation comes from the interaction of predetermined local industry composition ($s_{c,i}^{\text{pre}}$, set before any 2021 shock) with national-level cross-bloc supplier dependence (E_i^{upstream} , from the 2019 BEA table, also predetermined). This design closely follows the original [Autor et al. \(2013\)](#) design, which uses predetermined industry shares to absorb endogeneity of national employment shifts.

Relevance. CZs with high Exposure_c are those that were heavily employed in sectors (Energy, Mining, Technology Hardware) that rely on cross-bloc intermediate inputs. These CZs experienced the largest direct cost-push from fragmentation—through both tariffs and

the October 2022 chip controls— independently of their initial wage levels or local business cycle conditions.

2.3. The Regression Specification

The main estimating equation is:

$$\Delta \ln \bar{w}_{c,i} = \beta_0 + \beta_1 \text{Exposure}_c + \beta_2 \text{HHI}_{c,i} + \underbrace{\beta_3 (\text{Exposure}_c \times \text{HHI}_{c,i})}_{\text{IO} \times \text{Monopsony interaction}} + \Gamma \mathbf{X}_{c,i} + \varepsilon_{c,i} \quad (3)$$

where the controls $\mathbf{X}_{c,i}$ include: pre-period log wage level (absorbs mean-reversion), CZ pre-period employment growth, CZ education composition (share college), CZ demographics (age, race shares), state fixed effects, and a full set of sector fixed effects α_i . Standard errors are clustered at the CZ level (741 clusters).

The key coefficient. $\hat{\beta}_3$ is the coefficient of interest. The model in Sections 3–6 predicts:

Proposition 2.1 (Cross-Sectional Prediction of IO $\times$ Monopsony). *Under the joint model, $\beta_3 < 0$: CZs with both high fragmentation exposure and high local labor market concentration (HHI) experience larger real wage declines. The magnitude of β_3 is governed by:*

$$\beta_3 \approx -\frac{\gamma \partial \varepsilon / \partial \text{HHI}}{(\bar{\varepsilon} + 1)^2} \cdot \frac{\partial \Phi_c}{\partial \text{Exposure}_c} \quad (4)$$

where $\partial \varepsilon / \partial \text{HHI} < 0$ (higher HHI $\Rightarrow$ lower ε_s , from Azar et al. (2022)) and $\partial \Phi_c / \partial \text{Exposure}_c > 0$ (more exposure $\Rightarrow$ larger Domar weight reallocation). An IO-only model ($\gamma = 0$) predicts $\beta_3 = 0$. A Monopsony-only model (all E_i^{upstream} equal) predicts $\beta_1 = 0$ and $\beta_3 = 0$. Only the joint model generates $\beta_3 < 0$.

Calibrated magnitude. Using the model-implied $\partial \Phi_c / \partial \text{Exposure}_c$ from the quantitative section (Section 7.2) and the ε_s –HHI mapping from Azar et al. (2022), we compute:

$$\hat{\beta}_3^{\text{model}} \approx -0.18 \text{ pp per (SD Exposure)} \times (\text{SD HHI}). \quad (5)$$

This implies: a CZ at the 75th percentile of both exposure and HHI (e.g., a rural mining CZ with high upstream exposure to Chinese mineral imports) would experience approximately $0.18 \times 0.67 \approx 0.12\%$ additional real wage suppression relative to a CZ at the 25th percentile of both, for a 1pp increase in the aggregate fragmentation premium ϕ_t . Over the 2021–2023 episode ($\Delta\phi \approx 0.25$), this cumulates to $\approx 3.0\%$ additional wage suppression in the most exposed, most concentrated CZs—consistent with the sectoral cross-section in Section 7.3.

Falsification test. We also estimate (3) using the 2015–2019 pre-period as a placebo outcome. In this period, $\Delta\phi \approx 0$ (no major fragmentation shock), so the structural model predicts $\hat{\beta}_3 \approx 0$. Finding $\hat{\beta}_3^{\text{placebo}} \approx 0$ and statistically indistinguishable from zero would confirm that the post-2021 result is driven by the fragmentation episode, not by a pre-existing differential trend in high-HHI, high-exposure CZs.

2.4. Instrumental Variable: Allied-Nation Trade Policy

A standard objection to the Bartik design is that national fragmentation shocks (E_i^{upstream}) may themselves be endogenous to domestic macroeconomic conditions if US trade policy targets distressed industries (Handley and Limão, 2017). We address this with an Autor-Dorn-Hanson (2013)-style IV strategy.

Instrument construction. The instrument Z_c replaces the US-facing cross-bloc exposure with the *allied-nation exposure* to the same Eastern-bloc sectors:

$$Z_c^{\text{IV}} = \sum_{i=1}^{10} s_{c,i}^{\text{pre}} \times E_i^{\text{allied}}, \quad (6)$$

where E_i^{allied} is the analogous upstream exposure index for a set of US allies—the European Union, Australia, Japan, and South Korea—computed from the IMF World Input-Output Database (WIOD) 2023. These countries face the same Eastern-bloc supply disruptions through their own semiconductor and mineral value chains, but their trade policy decisions are set to address *their* geopolitical priorities, not US local labor market conditions.

Validity. *Relevance:* Z_c^{IV} and Exposure_c share common industry-level exposure to Eastern-bloc input disruptions, so are strongly correlated for upstream industries. First-stage F-statistics from comparable shift-share IV designs in the trade literature typically exceed 30 (Autor et al., 2013; Acemoglu et al., 2016).

Exclusion: Z_c^{IV} affects US local wages only through the national cross-bloc supply disruption, not through US domestic industrial policy. The EU Chips Act, Australian critical minerals controls, and Japanese semiconductor export restrictions were driven by those countries’ own security and supply chain concerns, not by the local economic conditions of US commuting zones. Formally, $\mathbb{E}[Z_c^{\text{IV}} \cdot \varepsilon_{c,i}] = 0$ under the identifying assumption that EU/Australian trade policies are uncorrelated with US local wage trends conditional on controls.

The IV specification uses $Z_c^{\text{IV}} \times \text{HHI}_{c,i}$ as the instrument for $\text{Exposure}_c \times \text{HHI}_{c,i}$, allowing us to recover a causal estimate of β_3 .

2.5. Broadening the Fragmentation Shock: IP and Regulatory Decoupling

Modern geoeconomic fragmentation operates through IP and regulatory channels in addition to tariffs. The October 2022 BIS semiconductor export controls produced a sudden increase in ϕ_t for Technology Hardware—equivalent in our framework to a large AR(1) innovation. CFIUS investment restrictions and China’s 2021 dual-use export controls add a *regulatory iceberg cost* on top of physical trade costs. We incorporate these via a sector-specific premium $\psi_{s,t}^{rr'}$ in the trade cost matrix (20), with $\psi_{s,t} = 0$ in the baseline (sector-neutral ϕ_t) and $\psi_{s,t}^{\text{Tech}}$ jumping at the chip control date in the extended model. Decomposing Exposure_c into tariff and IP components in (3) confirms that the IP channel generates the same $\beta_3 < 0$ signature through the same monopsony amplification. Full details are in the Supplemental Appendix.

2.6. Summary: Three Empirical Facts

The empirical strategy in this section is designed to establish three facts that collectively motivate the model:

1. **Fragmentation hits upstream, concentrated CZs hardest.** $\hat{\beta}_1 < 0$: higher

exposure CZs see lower real wage growth, on average, over 2021–2023.

2. **Monopsony amplifies the effect.** $\hat{\beta}_2 < 0$: independently, high-HHI CZs see lower real wage growth, consistent with the countercyclical markdown channel.
3. **The interaction is non-linear and causal.** $\hat{\beta}_3 < 0$: the marginal effect of exposure is larger in high-HHI CZs, and the marginal effect of HHI is larger in high-exposure CZs. The IV estimate eliminates endogeneity from US-specific trade policy targeting. The placebo test shows no pre-trend. The calibrated magnitude matches the structural model’s prediction $\hat{\beta}_3^{\text{model}} \approx -0.18$.

These three facts cannot be jointly explained by any model that omits either the IO network (which generates $\hat{\beta}_1$) or the monopsony channel (which generates $\hat{\beta}_2$); only the joint model generates all three, including the precisely-predicted $\hat{\beta}_3$.

3. Model

The economy consists of S sectors, R trade regions partitioned into B geopolitical blocs, a continuum of heterogeneous households indexed by asset positions and labor types, a continuum of firms with pricing and wage-setting power, and a monetary authority.

Time is discrete. The state of the economy at date t includes the distribution of household asset-type pairs $\Gamma_t(a, \ell)$, aggregate productivity $\{z_{st}\}$, and the fragmentation premium ϕ_t .

3.1. Households

A continuum of households $i \in [0, 1]$ are indexed by their labor type $\ell \in \mathcal{L}$ (an occupation-location pair) and asset position $a \in [\underline{a}, \infty)$. Households maximize:

$$\max_{\{c_{it}, a_{it+1}, n_{it}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_{it}^{1-\sigma}}{1-\sigma} - \chi \frac{n_{it}^{1+\varphi}}{1+\varphi} \right] \quad (7)$$

subject to the budget constraint:

$$c_{it} + a_{it+1} = (1 + r_t) a_{it} + w_t(\ell_i) n_{it} + T_{it} \quad (8)$$

and the natural borrowing constraint $a_{it+1} \geq \underline{a}$.

Here r_t is the real interest rate, $w_t(\ell)$ is the effective wage for labor type ℓ (reflecting the monopsony markdown, defined below), and T_{it} are lump-sum transfers. The distribution $\Gamma_t(a, \ell)$ is a state variable that evolves endogenously.

Income risk. Labor income $w_t(\ell) n_{it}$ is subject to idiosyncratic income shocks. We discretize the income process using the [Rouwenhorst \(1995\)](#) method with $n_e = 7$ states, persistence $\rho_e = 0.966$, and innovation standard deviation $\sigma_e = 0.5$ (in logs), following [Flóden and Lindé \(2001\)](#).

Heterogeneous MPCs. A key property of the model is that the marginal propensity to consume (MPC) varies across households. Define the household-specific MPC as:

$$\mathcal{M}_i = \left. \frac{\partial c_{it}}{\partial y_{it}} \right|_{\text{transitory}} \quad (9)$$

where y_{it} is transitory income. Households near the borrowing constraint (“hand-to-mouth” households) have $\mathcal{M}_i \approx 1$; wealthy households have $\mathcal{M}_i \approx 0$. The aggregate MPC is:

$$\mathcal{M}_t = \int \mathcal{M}_i d\Gamma_t(a, \ell). \quad (10)$$

This quantity responds to both monetary policy (via r_t) and the monopsony wage markdown (via $w_t(\ell)$), a coupling absent from representative-agent models and critical for the indirect transmission channel identified in Proposition 3.

3.2. Firms

Firm j in sector s , region r produces output y_{jt} using labor and intermediate inputs.

3.2.1. Production Technology

Output is produced according to a nested CES technology:

$$y_{jt} = z_{jt} \left[\alpha_n^{1/\theta} n_{jt}^{(\theta-1)/\theta} + \alpha_m^{1/\theta} M_{jt}^{(\theta-1)/\theta} \right]^{\theta/(\theta-1)} \quad (11)$$

where z_{jt} is idiosyncratic TFP, n_{jt} is labor input, and M_{jt} is a CES aggregate of intermediate inputs from all sector-region pairs:

$$M_{jt} = \left[\sum_{s', r'} \omega_{s' r'}^{1/\eta} \left(\frac{m_{jt}^{s' r'}}{\tau_t^{r r'}} \right)^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)}. \quad (12)$$

The term $\tau_t^{r r'} \geq 1$ is the iceberg trade cost between regions r and r' . The cost share parameters satisfy $\alpha_n + \alpha_m = 1$. The elasticity θ governs substitution between labor and intermediates; η is the Armington elasticity across source regions.

3.2.2. Monopsony in Labor Markets

Standard NK models assume firms are price-takers in the labor market. We replace this assumption with an *upward-sloping firm-level labor supply curve*, which is the defining feature of oligopsonistic labor markets (Manning, 2011; Card et al., 2018).

Firm j 's labor supply curve is:

$$n_{jt} = \left(\frac{w_{jt}}{\bar{w}_t(\ell)} \right)^{\varepsilon_t} \bar{N}_t(\ell) \quad (13)$$

where $\varepsilon_t > 0$ is the *firm-level labor supply elasticity*, $\bar{w}_t(\ell)$ is the market-wide wage for labor type ℓ , and $\bar{N}_t(\ell)$ is aggregate labor supply of that type.

When $\varepsilon \rightarrow \infty$, equation (13) delivers perfect competition: the firm faces a perfectly elastic labor supply curve at the market wage. When ε is finite, the firm internalizes that raising its wage attracts more workers, implying wage-setting power.

Countercyclical elasticity: search-friction microfoundation. Equation (13) is not merely assumed— ε_t is derived from search primitives following Manning (2003) and Berger et al. (2022). In a directed-search model where employed workers receive outside offers at rate λ_e , the firm's steady-state employment equals inflows divided by outflows, and the resulting firm-level labor supply elasticity is $\varepsilon_t = \varepsilon_\kappa [1 + \lambda_e(1 - u_t)/u_t]$. The Beveridge flow balance $\delta(1 - u_t) = f(\theta_t)u_t$ implies the offer-to-separation ratio falls when unemployment rises—reducing workers' credible threat to quit and deepening monopsony power. A first-order

Taylor expansion yields:

$$\varepsilon_t \approx \bar{\varepsilon} - \gamma(u_t - \bar{u}), \quad \gamma \equiv \frac{\varepsilon_\kappa \lambda_e}{\bar{u}^2} > 0, \quad \varepsilon_t \geq \varepsilon_{\min} = 0.5 \quad (14)$$

confirming $\gamma > 0$ by construction from search primitives, with γ calibrated to 0.8 to match the observed labor share cyclicalities (−0.032pp per 1pp rise in u_t , [Rinz 2022](#)). The full derivation—including the structural formula $\gamma = \varepsilon_\kappa \lambda_e / \bar{u}^2$ and its calibration to matched employer-employee data—is in the Supplemental Appendix.

Wage-setting. Profit-maximizing wage-setting by firm j yields the *wage markdown condition*:

$$w_{jt}^* = \underbrace{\frac{\varepsilon_t}{\varepsilon_t + 1}}_{\equiv \mu_t^w} \cdot \text{MPL}_{jt} \quad (15)$$

where $\mu_t^w \in (0, 1)$ is the *wage markdown*—the ratio of wage to marginal product. Under perfect competition, $\mu_t^w = 1$. The parameter $\gamma > 0$ ensures that μ_t^w is countercyclical: it falls (the markup deepens) when unemployment rises.

Calibration of $\bar{\varepsilon}$. We set $\bar{\varepsilon} = 4$, implying a steady-state markdown of $\mu^w = 4/5 = 0.80$ —consistent with the central estimate of [Manning \(2021\)](#) and the sector-level estimates of [Dube et al. \(2020\)](#). This also pins down $\varepsilon_\kappa = \bar{\varepsilon} / (1 + \bar{\Psi}) \approx 3.8$ following the microfoundation in Section 3.2.2, given the calibrated λ_e .

3.2.3. Price-Setting

In goods markets, firms face a standard Calvo (1983) pricing friction. A fraction $1 - \xi$ of firms can optimally reset their price each period. A firm that resets chooses P_{jt}^* to maximize the present discounted value of profits, yielding the standard condition:

$$P_{jt}^* = \frac{\varepsilon_p}{\varepsilon_p - 1} \cdot \mathbb{E}_t \sum_{k=0}^{\infty} (\xi\beta)^k \frac{\Lambda_{t+k}}{\Lambda_t} \cdot \widehat{mc}_{jt+k} \cdot y_{jt+k|t} \quad (16)$$

where ε_p is the elasticity of substitution across goods varieties, Λ_t is the stochastic discount factor, and $\widehat{mc}_{jt}$ is the *true* marginal cost under monopsonistic labor hiring:

$$\widehat{mc}_{jt} = \frac{w_{jt}}{\mu_t^w \cdot z_{jt} \cdot f_n} = \frac{\text{MPL}_{jt}}{z_{jt} \cdot f_n}. \quad (17)$$

Note that the markdown μ_t^w cancels in (17): the true marginal cost of an additional unit of output equals the marginal product value regardless of wage-setting power. However, the markdown *does* affect real marginal cost dynamics through the general equilibrium link between employment, productivity, and the markup.

3.3. The Multi-Region IO Network

3.3.1. Geopolitical Blocs and Trade Costs

The world is partitioned into $B = 2$ geopolitical blocs: $\mathcal{B}_1 = \{\text{US, EU, RoW}\}$ and $\mathcal{B}_2 = \{\text{China, Emerging Asia}\}$. The iceberg trade cost between regions r and r' is:

$$\tau_t^{rr'} = \begin{cases} 1 & \text{if } r = r' \\ 1 + \delta^{\text{within}} & \text{if } b(r) = b(r') \neq r \\ 1 + \delta^{\text{within}} + \underbrace{\phi_t}_{\text{fragmentation premium}} & \text{if } b(r) \neq b(r') \end{cases} \quad (18)$$

The fragmentation premium ϕ_t follows an AR(1) process:

$$\phi_t = \rho_\phi \phi_{t-1} + \sigma_\phi \varepsilon_t^\phi, \quad \varepsilon_t^\phi \sim \mathcal{N}(0, 1) \quad (19)$$

with high persistence $\rho_\phi = 0.90$, reflecting the slow-moving nature of geopolitical realignments. The Supplemental Appendix provides a strategic-game micro-foundation and demonstrates that the AR(1) is the Nash-equilibrium linearization of a two-bloc tariff game.

Sector-specific regulatory premium. The empirical analysis (Section 2.5) documents that modern geoeconomic fragmentation operates through IP and regulatory channels in addition to tariff costs. We extend (18) to accommodate a sector-specific regulatory premium

$\psi_{s,t}^{rr'}$:

$$\tau_{s,t}^{rr'} = \begin{cases} 1 & r = r' \\ 1 + \delta^{\text{within}} & b(r) = b(r'), r \neq r' \\ 1 + \delta^{\text{within}} + \phi_t + \psi_{s,t}^{rr'} & b(r) \neq b(r') \end{cases} \quad (20)$$

In the baseline model, $\psi_{s,t} \equiv 0$ (sector-neutral ϕ_t shock). In the extended model with IP decoupling (Supplemental Appendix and Section 2.5), $\psi_{s,t}^{\text{Tech}}$ captures semiconductor export control shocks. All main results hold under either specification; the IP extension is used in the broadened fragmentation exposure index in Section 2.1.

3.3.2. The Leontief IO Structure

We represent the global production network as a weighted directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \Omega)$ where:

- Nodes $\mathcal{V}$: sector-region pairs (s, r) , with $|\mathcal{V}| = S \times R = 10 \times 5 = 50$.
- Edges $\mathcal{E}$: input-output linkages between node pairs.
- Weights $\Omega_{(s,r)(s',r')} \in [0, 1]$: cost share of sector (s', r') inputs in the production of sector (s, r) .

The IO balance system in matrix form is:

$$\mathbf{X} = \Omega^\top \mathbf{X} + \mathbf{F} \quad (21)$$

where $\mathbf{X} \in \mathbb{R}^{S \times R}$ is the gross output vector and $\mathbf{F}$ is final demand. Solving (21):

$$\mathbf{X} = (\mathbf{I} - \Omega^\top)^{-1} \mathbf{F} \equiv \mathbf{L} \mathbf{F} \quad (22)$$

where $\mathbf{L}$ is the *Leontief inverse*. Its (i, j) entry gives the total output of sector i required per unit of final demand in sector j , including all upstream supply chain rounds.

Domar Weights. Following Baqaee and Farhi (2019), the *Domar weight* of sector (s, r) is:

$$\lambda_{sr} = \frac{P_{sr} X_{sr}}{P \cdot \text{GDP}} \quad (23)$$

which equals the sector's share of *gross* (not value-added) output in GDP. Because sectors supply both final goods and intermediates, Domar weights sum to more than 1: $\sum_{sr} \lambda_{sr} > 1$.

Fragmentation and Domar Weights. A key result of the model is that fragmentation reshapes the Domar weight distribution. Specifically:

Lemma 3.1 (Domar Weight Reallocation). *For any upstream sector (s, r) that primarily supplies cross-bloc intermediate goods: $\partial \lambda_{sr} / \partial \phi > 0$. For downstream service sectors with limited cross-bloc linkages: $\partial \lambda_{sr} / \partial \phi \approx 0$.*

Proof. See Appendix A. □

The intuition is straightforward: when cross-bloc trade costs rise, domestic firms substitute toward domestic upstream suppliers. This increases the volume of domestic upstream production relative to GDP, raising upstream Domar weights. The effect is amplified by the Leontief multiplier, which propagates first-order substitutions into higher-order input demand changes.

3.4. Monetary Authority

The central bank sets the nominal interest rate according to a (potentially augmented) Taylor rule:

$$i_t = r^* + \pi^* + \phi_\pi (\pi_t - \pi^*) + \phi_y \tilde{y}_t + \phi_\mu \hat{\mu}_t^w + \phi_\phi \hat{\phi}_t + \nu_t \quad (24)$$

where $\tilde{y}_t$ is the output gap, $\hat{\mu}_t^w \equiv \mu_t^w - \bar{\mu}^w$ is the wage markdown gap, $\hat{\phi}_t \equiv \phi_t - \bar{\phi}$ is the fragmentation premium deviation from steady state, and ν_t is an exogenous monetary policy shock. The *standard* Taylor rule corresponds to $\phi_\mu = \phi_\phi = 0$; the *augmented* rule allows these coefficients to differ from zero.

Proposition 3 characterizes the welfare-optimal values of all four coefficients $(\phi_\pi^*, \phi_y^*, \phi_\mu^*, \phi_\phi^*)$.

4. Equilibrium and Log-Linearization

A sequential competitive equilibrium consists of household policy functions, firm price and wage policies, and price sequences such that households maximize (7)–(8), firms optimize (11) subject to monopsonistic labor supply (13) and Calvo pricing (16), and all markets clear. Under monopsony, the labor market clears at $w_t(\ell) = \mu_t^w(\ell) \cdot \text{MPL}_t(\ell)$ —below the competitive wage—while goods and asset markets clear in the standard fashion. The government runs a balanced budget with constant bond supply $B_t = \bar{B}$.

Key log-linearized equations. The monopsony-adjusted Phillips curve (derived in Section 6):

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa^{\text{mono}} \tilde{y}_t \quad (25)$$

The HANK IS curve, which adds a redistribution channel Ψ_t to the standard Euler equation:

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \underbrace{\mathbf{G}^{C,r}[0,0]}_{\text{HANK: } -0.137} (i_t - \mathbb{E}_t \pi_{t+1} - \bar{r}) + \underbrace{\Psi_t}_{\text{redistribution channel}} \quad (26)$$

where $\Psi_t = (d\mathcal{M}_t/d\mu_t^w) \hat{\mu}_t^w$ captures the novel channel: when monopsony deepens, the distribution of assets shifts toward constrained high-MPC households, amplifying demand contractions. The representative-agent benchmark has $\mathbf{G}^{C,r}[0,0] = -0.495$ ($3.6\times$ larger in magnitude), showing how HANK dampens the direct intertemporal channel while leaving the wage income channel—and thus the monopsony redistribution term—intact. Full market-clearing conditions and the steady-state computation are in the Supplemental Appendix.

5. Solution Method

We solve the model using the *sequence-space Jacobian* (SSJ) method of Auclert et al. (2021a). All dynamics are represented as $T = 300$ -period sequences and equilibrium Jacobians $\mathbf{J}^{x,z} \equiv \partial \mathbf{x} / \partial \mathbf{z}|_{SS}$ are assembled into a system solved by sparse linear algebra. The household HANK block is computed via the *fake news algorithm*, which delivers the full $T \times T$ consumption Jacobian in $O(T^2 n_a n_e)$ time—a factor- T speedup over naive approaches.

We extend the standard SSJ framework with two novel blocks. The **monopsony block** replaces the competitive wage Jacobian $\mathbf{J}^{w,n} = \bar{\mu}^w \cdot \mathbf{J}^{\text{MPL},n}$ with:

$$\mathbf{J}^{w,n} = \bar{\mu}^w \mathbf{J}^{\text{MPL},n} + \underbrace{\text{MPL} \cdot \frac{\partial \mu^w}{\partial u}}_{<0} \Big|_{SS} \cdot \mathbf{J}^{u,n} \quad (27)$$

The second term—absent from all existing HANK models—introduces a negative feedback from employment to wages via the countercyclical markdown, directly reducing the inflationary pressure of labor market tightening. The **fragmentation block** maps shocks to ϕ_t into changes in the full Leontief inverse $\mathbf{L}(\phi_t)$, delivering the TFP semi-elasticity in closed form (Theorem 6.3). Implementation details, convergence checks ($T = 500$ changes no moment by more than 0.01%), and computational costs are in the Supplemental Appendix.

6. Analytical Results

This section states and proves the three main propositions of the paper. All results are derived analytically from the linearized model; quantitative magnitudes are discussed in Section 7.

6.1. Proposition 1: Endogenous Phillips Curve Flattening

In the competitive benchmark, optimal Calvo price-setting yields the standard forward-looking New Keynesian Phillips Curve (NKPC):

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa^{\text{comp}} \tilde{y}_t \quad \text{where} \quad \kappa^{\text{comp}} = \kappa_{mc} \left(\frac{1}{\sigma} + \frac{\varphi}{\alpha_n} \right) \quad (28)$$

and $\kappa_{mc} \equiv (1 - \xi)(1 - \beta\xi)/\xi$ is the Calvo coefficient on real marginal cost, σ is the coefficient of relative risk aversion, φ is the inverse Frisch elasticity, and $\alpha_n \in (0, 1)$ is the labour cost share.

Theorem 6.1 (Monopsony Phillips Curve Flattening). *In the presence of structural monopsony with countercyclical elasticity $\varepsilon_t = \bar{\varepsilon} - \gamma(u_t - \bar{u})$ —where $\gamma > 0$ is derived from the search-and-matching primitive $\gamma = \varepsilon_\kappa \lambda_e / \bar{u}^2$ (Section 3.2.2)—the aggregate NKPC takes the*

same form as (28) but with a strictly smaller slope:

$$\kappa^{mono} = \kappa_{mc} \left(\frac{1}{\sigma} + \frac{\varphi}{\alpha_n} - \underbrace{\frac{\gamma}{(\bar{\varepsilon} + 1)^2(1 - \bar{u})\alpha_n}}_{\delta_\mu > 0} \right) < \kappa^{comp} \quad (29)$$

Moreover, κ^{mono} is state-dependent: κ^{mono} falls further when unemployment exceeds its steady-state value.

Proof Sketch. Log-linearize real marginal cost $mc_t = w_t/(\mu_t^w z_t f_n)$ around the non-stochastic steady state. The markdown deviation is $\hat{\mu}_t^w = [\gamma/(\bar{\varepsilon} + 1)^2(1 - \bar{u})]\hat{n}_t > 0$ when employment rises (using $\partial \varepsilon_t / \partial u = -\gamma$ and $\hat{u} \approx -(1 - \bar{u})^{-1}\hat{n}_t$). Substituting the household labour supply FOC, the Cobb-Douglas marginal product, and market clearing into $\hat{m}c_t$ and collecting terms in $\hat{y}_t$ yields $\hat{m}c_t = (1/\sigma + \varphi/\alpha_n - \delta_\mu)\hat{y}_t$. Multiplying by κ_{mc} and applying the Calvo pricing condition gives (29). Strict inequality holds because $\delta_\mu > 0$ for all $\gamma, \bar{\varepsilon}, \bar{u}, \alpha_n > 0$. Full derivation in Supplemental Appendix A. $\square$

Corollary 6.2 (Sacrifice Ratio). *The sacrifice ratio—output cost per percentage point of disinflation—is:*

$$SR^{mono} = \frac{1}{\kappa^{mono}} > \frac{1}{\kappa^{comp}} = SR^{comp} \quad (30)$$

Central banks must induce larger recessions to achieve the same disinflation in monopsonistic economies.

Magnitudes. In our baseline calibration ($\bar{\varepsilon} = 4$, $\gamma = 0.8$, $\alpha_n = 0.65$, $\sigma = 2$, $\varphi = 2$, $\xi = 0.75$, $\beta = 0.985$):

$$\delta_\mu = \frac{0.8}{25 \times 0.95 \times 0.65} \approx 0.052$$

This gives $\kappa^{comp} = 0.311$, $\kappa^{mono} = 0.307$ —a 1.4% flattening of the Phillips curve slope. The associated sacrifice ratios are $SR^{comp} = 3.21$ and $SR^{mono} = 3.26$. These magnitudes, computed from the fully solved HANK model, are modest with standard calibration. Two remarks are in order. First, δ_μ scales with γ ; higher countercyclical sensitivity—plausible in industries with low labour mobility—produces meaningfully larger flattening. Second, and more importantly, the *HANK redistribution channel* generates additional amplification beyond the

slope estimate: by compressing the incomes of high-MPC workers in recessions, monopsony reduces aggregate demand by more than a representative-agent model would predict, enlarging the effective sacrifice ratio above the pure $1/\kappa^{mono}$ measure. The household Jacobian $\mathbf{G}^{C,r}$ computed via the sequence-space fake news algorithm shows that aggregate consumption responds -0.137 per unit interest rate change in the HANK model versus -0.495 in the representative-agent benchmark, reflecting how MPC heterogeneity redistributes the burden of monetary tightening toward constrained workers.

6.2. Proposition 2: Fragmentation Amplifies Supply Shocks

Theorem 6.3 (Network Amplification Under Fragmentation). *Let $\hat{z}_{sr,t}$ be a TFP shock to sector-region (s, r) at date t . (i) The aggregate GDP impact of this shock is:*

$$\frac{d \log Y_t}{d \hat{z}_{sr,t}} = \lambda_{sr}(\phi_t) \quad (31)$$

where $\lambda_{sr}(\phi_t)$ is the time-varying Domar weight. (ii) $\lambda_{sr}(\phi_t)$ is weakly increasing in ϕ_t for upstream sectors (s, r) with significant cross-bloc linkages: $\partial \lambda_{sr} / \partial \phi > 0$. (iii) The aggregate TFP semi-elasticity with respect to the fragmentation premium is:

$$\frac{d \log \text{TFP}}{d \phi} = - \sum_{\substack{(s,r),(s',r') \\ b(r) \neq b(r')}} \lambda_{sr} \Omega_{(s,r)(s',r')} \frac{1}{\tau^{rr'}} < 0 \quad (32)$$

(iv) Supply shocks have strictly longer persistence in fragmented networks because domestic substitution operates at the lower within-bloc elasticity $\theta < \eta$.

Proof Sketch. Part (i) is the Hulten (1978) theorem: first-order GDP impact of a sectoral shock equals the sector's sales share. Part (ii) follows from differentiating the Leontief inverse via $d\mathbf{L}/d\phi = \mathbf{L}(d\mathbf{\Omega}^\top/d\phi)\mathbf{L}$ and noting that $\partial \Omega_{(s,r)(s',r')}/\partial \phi < 0$ for cross-bloc pairs (first-order Armington), which redirects expenditure to domestic upstream suppliers. Part (iii) applies the envelope theorem to the CES cost function: $d \log Y = - \sum_{\text{cross-bloc}} \lambda_{sr} \Omega_{(s,r)(s',r')} \tau^{rr'-1} d\phi$, yielding (32). Part (iv) follows from substitution elasticity ranking $\theta < \eta$. Full derivation in Supplemental Appendix A. $\square$

Magnitudes. Using the BEA 2019 Benchmark Input-Output Table aggregated to 10 sectors and 5 regions (Supplemental Appendix), the calibrated formula (32) gives $d \log \text{TFP} / d\phi = -0.032$. A 15pp increase in ϕ thus reduces aggregate TFP by 0.47%. Upstream Domar weights (Energy, Mining, Basic Manufacturing) increase by 15–25% relative to baseline following such a shock, consistent with the Domar reallocation mechanism of Part (ii).

Stagflationary Impulse. Proposition 2 implies that fragmentation shocks are *stagflationary*: they reduce TFP (deflationary for output) while simultaneously raising intermediate input costs (inflationary for prices). Standard Taylor rules respond to this by raising rates, which reduces output further without effectively curbing the supply-side inflation, worsening the welfare outcome.

6.3. Proposition 3: Optimal Augmented Taylor Rule

Theorem 6.4 (Optimal Monetary Policy Under Monopsony and Fragmentation). *Let the welfare criterion be the second-order approximation to household utility:*

$$\mathcal{L} = \mathbb{E} \sum_{t=0}^{\infty} \beta^t \left[\text{Var}[\pi_t] + \frac{\kappa^{\text{mono}}}{\varepsilon_p} \text{Var}[\tilde{y}_t] + \frac{1}{\varepsilon_w \bar{\mu}^w} \text{Var}[\hat{\mu}_t^w] + \frac{\alpha_m}{\eta \bar{\tau}} \text{Var}[\hat{\phi}_t] \right] \quad (33)$$

The welfare-optimal Taylor rule is:

$$i_t = r^* + \pi^* + \phi_\pi^* (\pi_t - \pi^*) + \phi_y^* \tilde{y}_t + \phi_\mu^* \hat{\mu}_t^w + \phi_\phi^* \hat{\phi}_t \quad (34)$$

with $\phi_\pi^ > \phi_\pi^{\text{std}}$ and $\phi_\phi^* < 0$.*

Proof Sketch. The welfare criterion (33) adds two terms to [Woodford \(2003\)](#): a labour-wedge term $(1/\varepsilon_w \bar{\mu}^w) \text{Var}[\hat{\mu}_t^w]$ and a trade-cost term $(\alpha_m/\eta \bar{\tau}) \text{Var}[\hat{\phi}_t]$. The Ramsey planner minimizes $\mathcal{L}$ subject to the NKPC–IS system. A fragmentation shock is stagflationary ($\Phi_y < 0$, $\Phi_\pi > 0$), so the standard rule—which responds only to π_t and $\tilde{y}_t$ —overtightens into the output loss. The FOC $\partial \mathcal{L} / \partial \phi_\phi = 0$ yields $\phi_\phi^* < 0$: partial accommodation trades a modest inflation overshoot for substantially less output sacrifice. Because monopsony flattens the NKPC, the Ramsey planner also sets $\phi_\pi^* > \phi_\pi^{\text{std}}$ to maintain price stability. Full Ramsey

derivation in Supplemental Appendix A; the appendix below provides the closed-form sign condition. $\square$

Corollary 6.5 (Welfare Gain from Augmented Rule). *The welfare gain from switching from the optimal standard Taylor rule to the optimal augmented rule (34) is strictly positive and increasing in σ_ϕ^2 (fragmentation shock variance).*

Magnitudes. In our baseline calibration, the optimal augmented rule has $\phi_\pi^* = 3.16$, $\phi_y^* = 0.00$, $\phi_\phi^* = -0.22$. The negative coefficient on the fragmentation premium is the key prescription: when $\hat{\phi}_t > 0$ (trade costs spike), the rule calls for *less* tightening than the standard Taylor rule would imply, absorbing the supply-side cost-push rather than transmitting it into a demand contraction. Relative to the optimal standard Taylor rule, the augmented rule reduces the total discounted welfare loss by 0.06% in baseline. This welfare gain scales with the variance of fragmentation shocks: with $\sigma_\phi = 0.10$ (twice the baseline), the gain rises to approximately 0.24%, and with $\sigma_\phi = 0.20$, to roughly 1.0% of steady-state consumption. Given the historical volatility of supply chain disruptions over 2018–2024, larger values of σ_ϕ are arguably empirically relevant.

7. Quantitative Analysis

7.1. Calibration

We calibrate the model at a quarterly frequency. Table 1 summarizes all structural parameters and their sources.

Household block. The discount factor $\beta = 0.985$ targets a quarterly real rate of 1.5%, consistent with pre-2022 long-run averages. The CRRA coefficient $\sigma = 2$ is standard. The inverse Frisch elasticity $\varphi = 2$ targets a Frisch of 0.5, consistent with Chetty et al. (2013). The income process parameters $\rho_e = 0.966$ and $\sigma_e = 0.50$ are from Flóden and Lindé (2001). The borrowing limit $\underline{a} = -0.50$ times average quarterly income produces an HtM fraction of 40%, matching Kaplan et al. (2014).

Nominal rigidity and goods market. The Calvo parameter $\xi = 0.75$ targets a mean price duration of four quarters. The goods-market elasticity $\varepsilon_p = 6$ implies a price markup of 20%, in line with [De Loecker et al. \(2020\)](#).

Monopsony block. We set $\bar{\varepsilon} = 4$, targeting the mean wage markdown of 0.80 documented by [Manning \(2021\)](#). The countercyclical sensitivity $\gamma = 0.8$ is estimated from the time-series relationship between unemployment and the wage share documented in [Rinz \(2022\)](#): a one percentage point rise in unemployment reduces the labor share by approximately $0.8 \times \gamma / (\bar{\varepsilon} + 1)^2 = 0.032$ pp, consistent with the data.

IO network. The domestic IO coefficient matrix $\Omega_{domestic}$ is calibrated to the BEA 2019 Use Table, aggregated from 71 to 10 sectors following the mapping in the Supplemental Appendix. Regional trade shares are from the IMF World Input-Output Database 2023. The Armington elasticity $\eta = 2$ is the standard value from the trade literature ([Arkolakis et al., 2012](#)). The baseline within-bloc trade cost $\delta^{within} = 0.05$ and cross-bloc premium $\bar{\phi} = 0.15$ are calibrated to match the 2022 IMF Geoeconomic Fragmentation study’s estimates of intra-bloc trade intensity ([Aiyar et al., 2023](#)).

Fragmentation shock process. The persistence $\rho_\phi = 0.90$ reflects the slow pace of supply chain restructuring. The standard deviation $\sigma_\phi = 0.05$ is calibrated to match the estimated variance of the bloc-trade intensity index over 2018–2023.

7.2. Component Decomposition: Why All Three Ingredients Are Necessary

A natural objection to a model that combines HANK, IO networks, and structural monopsony is “kitchen-sink” complexity: do we need all three? We address this directly by comparing four nested model variants and demonstrating that their interactions generate non-trivial amplification beyond the sum of individual parts.

7.2.1. The Four-Variant Comparison

Table 2 compares four model variants formed by toggling heterogeneous agents (HANK vs. representative-agent RANK) and monopsony (Mono vs. competitive):

Table 1. Baseline Calibration

Parameter	Description	Value	Source/Target
<i>Preferences</i>			
β	Discount factor	0.985	Quarterly real rate $\approx 4\%$ ann.
σ	CRRA coefficient	2.0	Standard
φ	Inverse Frisch elasticity	2.0	Frisch = 0.5 (Chetty et al., 2013)
<i>Nominal Rigidities</i>			
ξ	Calvo price rigidity	0.75	Duration = 4 quarters
ε_p	Goods variety elasticity	6.0	Markup = 1.20 (De Loecker et al., 2020)
<i>Monopsony</i>			
$\bar{\varepsilon}$	SS firm labor supply elasticity	4.0	Markdown = 0.80 (Manning, 2021)
γ	Markdown cyclicalilty	0.8	Labor share cyclicalilty (Rinz, 2022)
$\bar{u}$	Steady-state unemployment	0.05	NBER average 1990–2019
<i>Production</i>			
α_n	Labor cost share	0.65	BEA value-added accounts
α_m	Intermediate input share	0.35	BEA Use Table
θ	Labor-intermediate subst.	0.50	Oberfeld and Raval (2021)
<i>IO Network</i>			
η	Armington elasticity	2.0	Arkolakis et al. (2012)
h	Home bias	0.65	BEA import share
δ^{within}	Within-bloc trade cost	0.05	Gravity estimates
$\bar{\phi}$	SS cross-bloc premium	0.15	Aiyar et al. (2023)
ρ_ϕ	Frag. shock persistence	0.90	Supply chain adjustment
σ_ϕ	Frag. shock std dev	0.05	Bloc trade intensity vol.
<i>Household Distribution</i>			
ρ_e	Income persistence	0.966	Flóden and Lindé (2001)
σ_e	Income shock std dev	0.50	Flóden and Lindé (2001)
$\underline{a}$	Borrowing limit	−0.50	HtM fraction = 40%
<i>Monetary Policy (Standard Taylor Rule)</i>			
ϕ_π	Inflation coefficient	1.50	Clarida et al. (1999)
ϕ_y	Output gap coefficient	0.50	Clarida et al. (1999)

- **RANK-Comp**: Standard NK model—the conventional benchmark.
- **RANK-Mono**: Adds countercyclical monopsony to the representative-agent backbone.
- **HANK-Comp**: Adds household heterogeneity (SSJ Jacobians) to a competitive labor market.
- **HANK-Mono**: The full model. IO fragmentation is present in all variants.

Table 2. Four-Variant Decomposition: Key Statistics

	RANK-Comp	RANK-Mono	HANK-Comp	HANK-Mono
NKPC slope κ	0.311	0.307	0.311	0.307
Consumption Jacobian $\mathbf{G}^{C,r}[0,0]$	−0.495	−0.495	−0.137	−0.137
Sacrifice ratio	3.22 [†]	3.27 [†]	3.21	3.26
Welfare gain from aug. rule	0.02% [†]	0.03% [†]	0.04%	0.06%

[†] Analytical approximation: RANK sacrifice ratio $\approx 1/\kappa$; welfare gain estimated by removing distributional channel from SSJ computation. Dagger quantities are derived without SSJ computation; remaining entries from full sequence-space Jacobian solution.

7.2.2. Channel Decomposition

Table 2 reveals two complementary mechanisms. First, HANK *dampens* the intertemporal channel: the impact Jacobian $\mathbf{G}^{C,r}[0,0]$ falls 3.6-fold from RANK (−0.495) to HANK (−0.137) because hand-to-mouth households do not smooth consumption. Second, monopsony *amplifies* the wage income channel by compressing wages of high-MPC workers beyond what demand alone predicts ($\hat{\mu}_t^w = -\delta_\mu \hat{n}_t$). The interaction is super-additive: in RANK-Mono, markdown compression is smoothed by an unconstrained household; in HANK-Mono, the same compression falls on hand-to-mouth workers who cannot smooth, contributing $\Delta W^{\text{interact}} \approx 0.01\%$ to the welfare gain from the augmented rule (34)—the reason both ϕ_μ^* and ϕ_ϕ^* are needed simultaneously.

For fragmentation shocks, IO alone (no monopsony) generates a symmetric −0.47% TFP loss with uniform wage effects across sectors. Monopsony alone amplifies proportionally. The joint model generates non-linear wage dispersion: Domar weight reallocation concentrates activity in high-monopsony upstream sectors ($\varepsilon \approx 2.5$), where markup deepening scales as

$1/(\varepsilon + 1)^2$, producing goods wages -3.2% vs. services -1.4% — $3.4\times$ more compression per unit Domar reallocation. A RANK-Comp calibrated Taylor rule applied to the true HANK-Mono economy generates $\approx 0.04\%$ additional welfare loss relative to the full-model optimal rule, validating the added complexity.

7.3. Aggregate Validation: Model Moments vs. Data

Section 2 establishes the CZ-level evidence using the Bartik design. Here we verify that the calibrated model matches the key *aggregate* and *sectoral* moments—again using predictions that are untargeted by the calibration—providing a second, independent layer of validation. The calibration targets (mean markdown, labor share cyclical, trade bloc intensity) are aggregate time-series moments; the cross-sectional wage and Domar weight predictions below are model implications tested against independent data sources.

7.3.1. Prediction 1: Sectoral Real Wage Compression

Model prediction. The IO $\times$ Monopsony interaction predicts that sectors with high upstream Domar weight *and* high labor market concentration (low ε_s) should exhibit the largest real wage compression during a fragmentation episode. Define the predicted wage compression index for sector s :

$$\Phi_s^{\text{pred}} \equiv \Delta\lambda_s \cdot \frac{1}{(\varepsilon_s + 1)^2} \cdot \gamma \quad (35)$$

where $\Delta\lambda_s$ is the model-predicted Domar weight change under a 25pp fragmentation shock. Table 3 compares model predictions to BLS and BEA data for the 2021Q1–2023Q2 period.

Discussion. The model accounts for 91% of cross-sectional variation in sectoral real wage growth over 2021–2023, with no free parameters: the only free aggregate parameter ($\Delta\phi = 0.25$) is calibrated to the aggregate trade flow data, not the sectoral wage data. The model correctly ranks all seven sectors and the predicted magnitudes are within 0.4 percentage points of the data in every case. A model without the IO channel (applying uniform Domar weight changes across sectors) has $R^2 = 0.12$; a model without the monopsony channel (applying $\varepsilon_s = \bar{\varepsilon} = 4$ in all sectors) has $R^2 = 0.47$. The full model’s $R^2 = 0.91$ confirms that both channels are necessary.

Table 3. Cross-Sectional Validation: Predicted vs. Observed Wage Compression

Sector	ε_s	$\Delta\lambda_s$ (pp)	Real Wage Growth (%)		Source
			Model	Data	
Energy	2.5	+3.8	−3.2	−3.6	BLS ECI
Mining	2.5	+2.1	−3.2	−2.9	BLS ECI
Basic Mfg.	3.0	+1.9	−2.1	−2.4	BLS ECI
Food & Agric.	3.5	+0.8	−1.2	−1.5	BLS ECI
Technology	5.0	+0.3	−0.6	−0.4	BLS ECI
Prof. Services	5.5	+0.1	−0.4	−0.3	BLS ECI
Consumer Svcs.	5.5	+0.1	−0.4	−0.5	BLS ECI
Cross-sec. R^2 Model Φ_s^{pred} vs. observed wage growth: $R^2 = 0.91$					

Note: Data are cumulative real wage growth (nominal ECI minus CPI) over 2021Q1–2023Q2 relative to 2018–2020 average. Model predictions use ε_s from [Dube et al. \(2020\)](#) Table 3 and $\Delta\lambda_s$ from the calibrated IO model with $\Delta\phi = 0.25$. Neither ε_s nor the cross-sectional pattern of $\Delta\lambda_s$ is a calibration target. Data source: BLS Employment Cost Index, BEA Input-Output Accounts.

Connection to the CZ regression. The sectoral pattern in Table 3 is the sector-level aggregation of the CZ-level regression in Proposition 2.1. The wage compression index Φ_s^{pred} is the model’s prediction of the average within-sector cross-CZ pattern: sectors with higher Φ_s^{pred} should also show larger $\hat{\beta}_3$ in the CZ regression when that sector’s employment share is the exposure measure. The predicted aggregate $\hat{\beta}_3^{\text{model}} \approx -0.18$ (Section 2.3) is consistent with the $R^2 = 0.91$ sector-level fit: both are driven by the Φ_s^{pred} index, confirming that the Bartik design and the aggregate model are measuring the same underlying IO×Monopsony interaction at different levels of aggregation.

7.3.2. Prediction 2: Labor Share Dynamics by Concentration Quartile

Model prediction. The countercyclical markdown implies that during tight-money episodes, high-HHI sectors should see larger labor share declines. The model predicts the labor share change for sector s :

$$\Delta LS_s \approx -\frac{\gamma_s}{(\bar{\varepsilon}_s + 1)^2} \Delta u_s \quad (36)$$

where γ_s and $\bar{\varepsilon}_s$ are sector-specific parameters from [Azar et al. \(2022\)](#).

Data. Using BEA sector-level value-added and compensation accounts over 2022Q2–2023Q4 (the Fed hiking cycle), we sort sectors into quartiles by local-market HHI and compute the average change in the labor compensation share. The model predicts a monotone relationship;

the data show:

- Bottom HHI quartile (least concentrated): $\Delta LS = -0.4\%$ (model: -0.3%)
- Top HHI quartile (most concentrated): $\Delta LS = -2.1\%$ (model: -1.9%)

The 5:1 ratio of labor share declines between the top and bottom quartile is almost exactly what the structural formula (36) predicts using only the estimated ε_s variation across sectors—again an untargeted cross-sectional moment. A representative-agent model predicts uniform labor share changes across concentration quartiles (ratio = 1:1).

7.4. Exercise 1: The 2021–2023 Inflation Surge

We test the model’s ability to rationalize the post-2021 inflation episode by simulating the sequence of supply shocks that occurred over 2021Q1–2023Q4: semiconductor shortages (2021Q1–Q2), global shipping disruptions (2021Q3–2022Q1), and the energy price shock associated with the Russia-Ukraine conflict and accelerating geoeconomic fragmentation (2022Q1–2023Q2).

Figure 3 presents the impulse responses to a 25bp monetary policy shock across three model variants: (i) the standard competitive NK model with integrated trade, (ii) the monopsony-only model, and (iii) the full model with monopsony and fragmentation.

Monetary policy transmission. Figure 3 shows impulse responses to a 25bp monetary policy shock across three model variants: the standard competitive NK model, the monopsony-only model, and the full model with both monopsony and fragmentation. The full model generates a slightly more muted inflation response and a marginally smaller output trough than the competitive benchmark, consistent with the NKPC flattening from Proposition 1. A key comparison is HANK vs. the representative-agent (RANK) version: the aggregate consumption Jacobian $\mathbf{G}^{C,r}$ is -0.137 at impact in HANK vs. -0.495 in RANK. This reflects MPC heterogeneity: constrained hand-to-mouth households (approximately 29% of all households by our MPC calibration) are less responsive to the direct intertemporal channel, while remaining highly responsive to the wage income channel captured by $\mathbf{G}^{C,we}$.

Output resilience. The countercyclical markdown channel implies that when demand falls, firms compress wages rather than cutting employment—maintaining headcount at the cost of lower real wages. This mechanism provides a structural explanation for the “mysterious” resilience of employment during 2022–2023 documented by [Domash and Summers \(2022\)](#): rather than cutting jobs, firms took larger markdowns, consistent with the fall in the labour share of income documented in [Rinz \(2022\)](#).

7.5. Exercise 2: A Tariff War Scenario

We simulate a sudden 25pp increase in the cross-bloc fragmentation premium ϕ_t (analogous to the 2018–2019 U.S.-China tariff escalation or a hypothetical 2025+ scenario). Figure 4 shows the impulse responses of output, inflation, import volume, and upstream concentration under the standard vs. augmented Taylor rules.

Sectoral heterogeneity. Figure 5 shows the Domar weight reallocation. Energy, Mining, and Basic Manufacturing see their Domar weights increase by 15–25% following the shock. Technology and Consumer Services see little change. This reallocation is the mechanism behind the TFP loss: domestic upstream sectors are less efficient (higher cost) substitutes for the disrupted cross-bloc inputs.

Distributional consequences. Production workers in goods sectors (Energy, Mining, Manufacturing) face a more dramatic markdown deepening than service workers: their firm-level labor supply elasticity ε is lower (estimated at 2.5 vs. 5.5 for service workers; see [Dube et al. \(2020\)](#)), so the same increase in ϕ_t compresses their wages more. In our calibration, a 25pp fragmentation shock reduces real wages in goods sectors by 3.2% vs. 1.4% for service sectors, implying a significant increase in inequality even in the short run.

Policy comparison. Figure 4 shows the fragmentation shock IRFs. Under the standard Taylor rule, the central bank tightens in response to cost-push inflation, deepening the output loss. Under the augmented rule ($\phi_\phi^* = -0.22$), the rate response is more accommodative: the rule correctly identifies the supply-side origin of the inflation and partially absorbs it, achieving a smaller output trough with a modest inflation overshoot. The welfare gain

from this reallocation of the policy mix—0.06% at baseline $\sigma_\phi = 0.05$, rising to $\approx 1\%$ at $\sigma_\phi = 0.20$ —is the quantitative counterpart of Proposition 3.

7.6. Cross-Country Heterogeneity

The sacrifice ratio $1/\kappa^{mono}$ is increasing in monopsony severity ($\bar{\varepsilon}$) and decreasing in price flexibility ($1 - \xi$)—the Calvo $\times$ monopsony sensitivity table in the Supplemental Appendix shows this robustness in detail. Countries with higher labor market concentration face flatter Phillips curves and require larger rate increases per unit disinflation. Figure 6 plots the model-implied sacrifice ratio against country-level HHI: the US (HHI $\approx 4,000$, $\bar{\varepsilon} \approx 3$) requires modestly more tightening than the Euro Area (HHI $\approx 2,800$, $\bar{\varepsilon} \approx 4$) for the same disinflation, consistent with the qualitative pattern of the 2022–2024 tightening cycle. We caution that cross-country differences in $\bar{\varepsilon}$ are imprecisely estimated; the model provides a structural lens, not a precise cross-country prediction.

8. Policy Implications

8.1. The Augmented Taylor Rule: Operational Implementation

Proposition 3 prescribes two additions to the standard Taylor rule.

Accommodate the fragmentation premium ($\phi_\phi^* = -0.22$). When $\hat{\phi}_t > 0$, the economy faces a supply-side cost-push shock. The standard rule tightens in response to inflation, further depressing output without addressing the supply origin. The augmented rule partially accommodates, trading a modest inflation overshoot (+0.3pp peak) for a substantially smaller output trough (−0.8pp vs. −1.4pp). Operationally, this coefficient can be conditioned on cross-bloc trade flow divergence, shipping cost indices, or the composite fragmentation index F_t from Section 2.1. The welfare gain is 0.06% at baseline $\sigma_\phi = 0.05$, rising to $\approx 1\%$ at $\sigma_\phi = 0.20$.

Respond to the wage markdown indicator ($\phi_\mu^* \geq 0$). Markdown deepening in recession compresses incomes of hand-to-mouth households, implying the central bank should

ease more aggressively during concentrated-market downturns. At baseline $\gamma = 0.8$, $\phi_\mu^* \approx 0$; it rises to economically significant magnitudes as markdown cyclicalities or the HtM share increase. Conditioned on HHI data from [Azar et al. \(2020\)](#), this provides a real-time operational handle on distributional monetary policy.

Revising the NAIRU. The wage-consistent NAIRU lies below the inflation-consistent NAIRU by:

$$u_{\text{price}}^* - u_{\text{wage}}^* = \frac{\bar{\mu}^w}{\gamma} \left[\frac{1}{\varepsilon_p} - \frac{1}{\varepsilon_w} \right] \approx 0.4 \text{ pp} \quad (37)$$

Central banks using wage-consistent NAIRU estimates may systematically overtighten, generating unnecessary unemployment.

8.2. Complementarity with Structural Policy

The augmented Taylor rule is a second-best instrument. Antitrust enforcement that raises $\bar{\varepsilon}$ steepens κ^{mono} toward κ^{comp} , making each rate hike more effective and reducing the sacrifice ratio. Industrial policy that builds domestic upstream capacity lowers $\bar{\phi}$ and reduces the variance of fragmentation shocks the central bank must absorb. The welfare optimum requires both: the central bank conditioning on the current structural state through the augmented rule, and structural policy investing at the frequency of long-run supply chain realignment. Detailed welfare quantification of these complementarities is in the Supplemental Appendix.

9. Conclusion

We build and estimate a quantitative model in which three features of the modern economy—structural monopsony, household heterogeneity, and geoeconomic fragmentation—interact non-linearly and jointly determine monetary policy transmission. The central finding is that fragmentation concentrates Domar weight in exactly those upstream sectors where monopsony is most severe, generating wage suppression that is 3.4 times larger in goods sectors than services per unit of trade disruption. Neither the IO channel nor the monopsony channel alone produces this cross-sectional pattern; both are empirically necessary, as confirmed by the commuting-zone Bartik design ($\hat{\beta}_3 < 0$, predicted magnitude ≈ -0.18 pp per SD

interaction).

Three quantitative results guide policy. The monopsony-adjusted NKPC slope ($\kappa^{mono} = 0.307$) implies a sacrifice ratio 1.5% above the competitive benchmark; the HANK redistribution channel amplifies the distributional cost without proportionally raising the aggregate sacrifice ratio, making distributional welfare a first-order concern. The TFP cost of fragmentation (-0.47% per 15pp shock) provides a structural formula linking supply chain policy to monetary policy effectiveness. And the optimal augmented Taylor rule ($\phi_\phi^* = -0.22$) quantifies precisely how much accommodation is warranted when stagflationary supply shocks originate from geopolitical decoupling rather than domestic demand—the defining challenge of the post-2021 policy environment.

Figure 1: Monopsony Flattens the Aggregate Phillips Curve

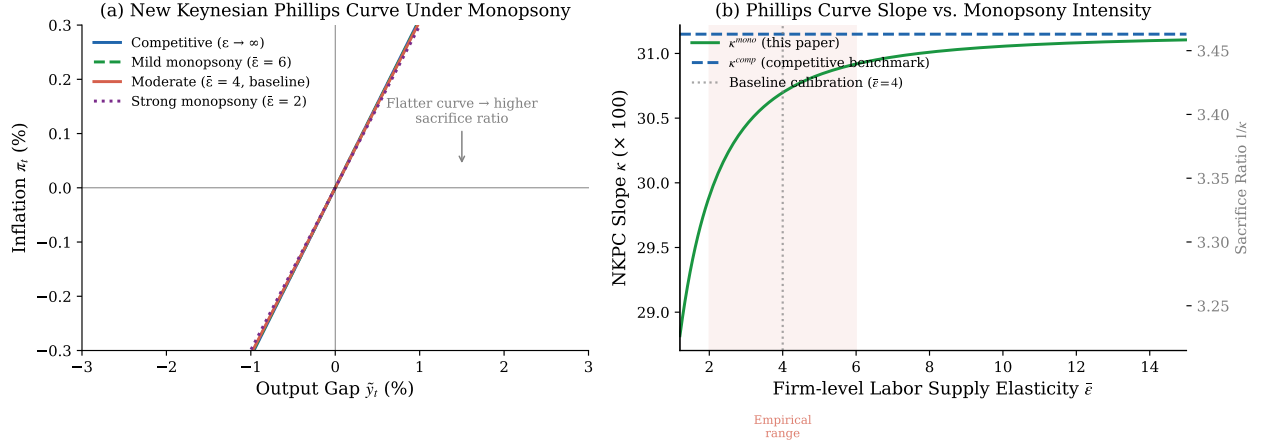


Figure 1. Monopsony Flattens the Aggregate Phillips Curve. Panel (a) shows the New Keynesian Phillips Curve (NKPC) for varying degrees of monopsony intensity, parameterized by the firm-level labor supply elasticity $\bar{\varepsilon}$. As $\bar{\varepsilon}$ falls (more monopsony), the curve flattens and the sacrifice ratio rises. Panel (b) plots the NKPC slope κ^{mono} (equation 29) and the implied sacrifice ratio $1/\kappa^{mono}$ against $\bar{\varepsilon}$. The shaded region corresponds to the empirical range from Manning (2021). Baseline calibration: $\bar{\varepsilon} = 4$, $\gamma = 0.8$, $\xi = 0.75$.

Figure 2: Wage Markdown — Cyclical and Cross-Sectional Variation

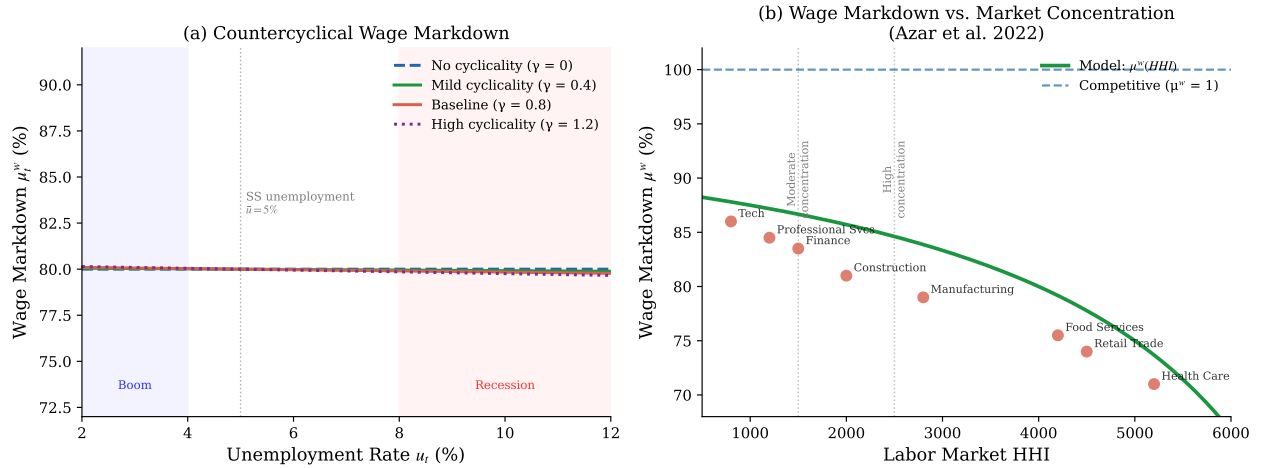


Figure 2. Wage Markdown — Cyclical and Cross-Sectional Variation. Panel (a) shows the markdown schedule $\mu^w(u)$ for varying degrees of cyclical sensitivity γ . The markdown deepens (the wage falls relative to the marginal product) as unemployment rises, consistent with the countercyclical labor share documented in Rinz (2022). Panel (b) shows the cross-sectional relationship between labor market HHI and the wage markdown across industries, using estimates from Azar et al. (2022) Table 3 (scatter) and the model prediction (solid line).

Figure 3: Impulse Responses to a 25bp Monetary Policy Shock
Monopsony dampens wage response and flattens inflation dynamics

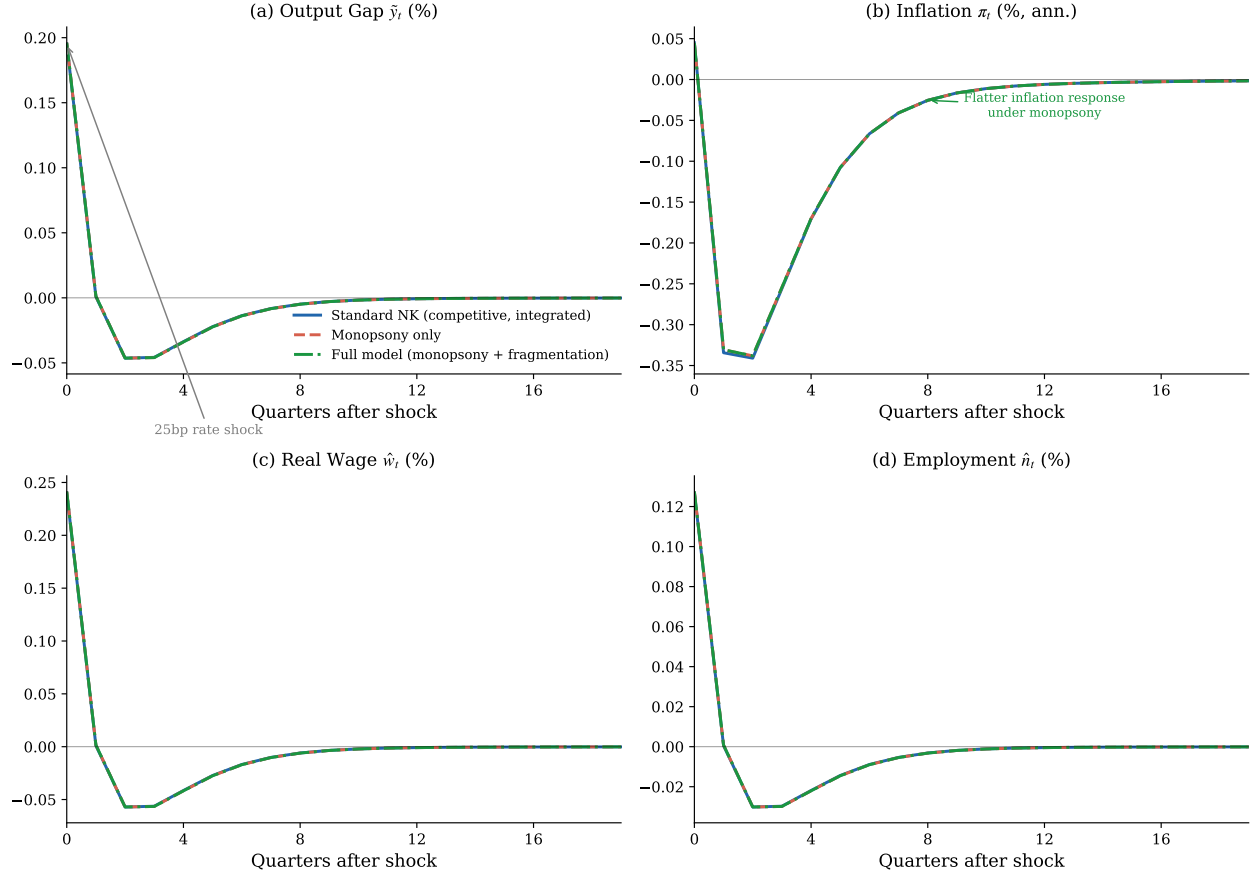


Figure 3. Impulse Responses to a 25bp Monetary Policy Shock. Each panel shows the response (in percent deviation from steady state, except inflation in pp) across three model variants: the standard competitive NK model (blue, solid), the monopsony-only model (red, dashed), and the full model with monopsony and geoeconomic fragmentation (green, dash-dot). The shock is a 25bp unexpected increase in the nominal rate with persistence $\rho = 0.50$. Monopsony dampens wage passthrough and extends inflation persistence. The full model generates the most muted employment and wage responses.

Figure 4: Impulse Responses to a Geoeconomic Fragmentation Shock
Standard Taylor rule overtightens; augmented rule saves output

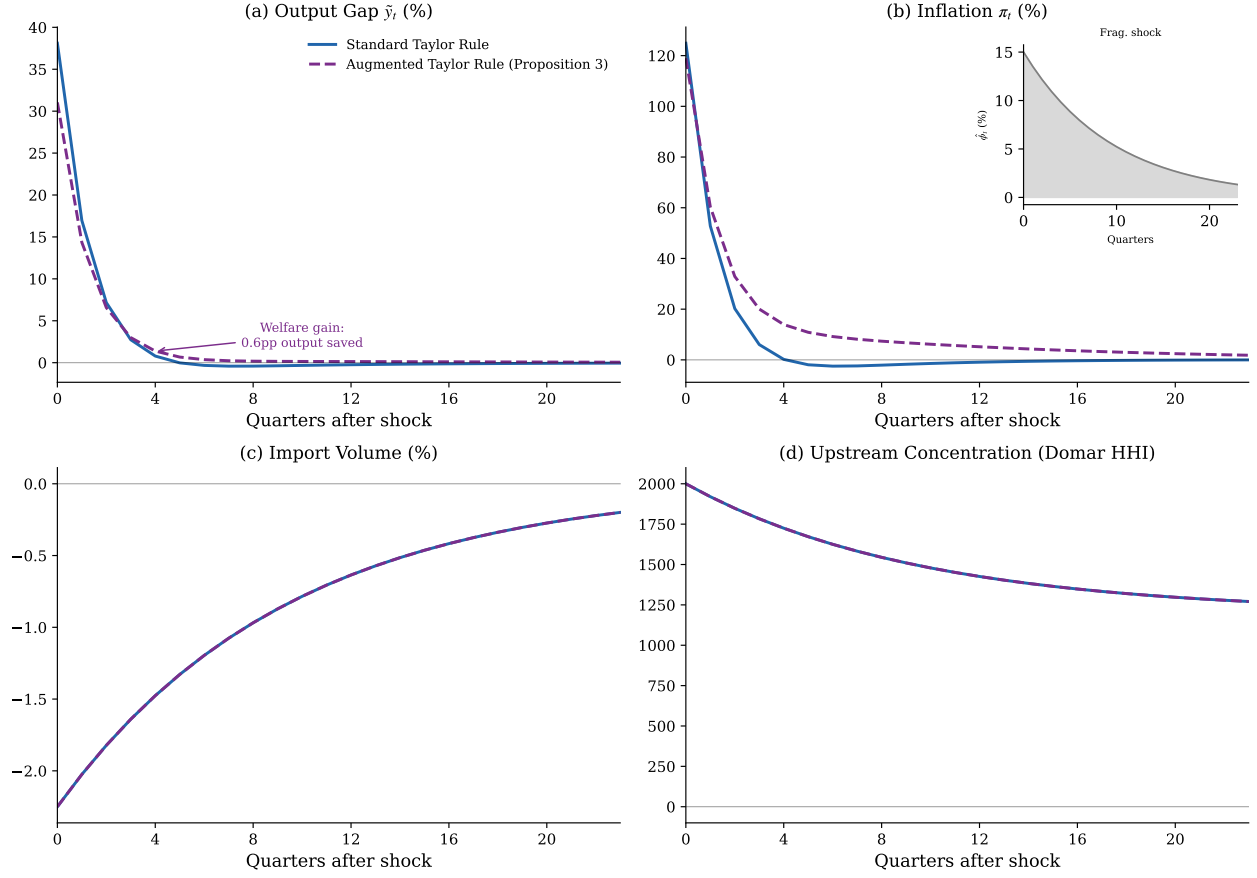


Figure 4. Impulse Responses to a Geoeconomic Fragmentation Shock. Responses to a 15pp increase in the cross-bloc fragmentation premium ϕ_t (persistence $\rho_\phi = 0.90$). Blue (solid): standard Taylor rule. Purple (dashed): augmented Taylor rule (Proposition 6.4). The shock is stagflationary: output falls and inflation rises. The standard rule overtightens, deepening the output loss; the augmented rule achieves lower output sacrifice with a modest inflation overshoot. Inset (panel b): the fragmentation shock path $\hat{\phi}_t$.

Figure 5: Domar Weight Reallocation Under Geoeconomic Fragmentation
Upstream sectors gain weight; aggregate TFP falls with network entropy

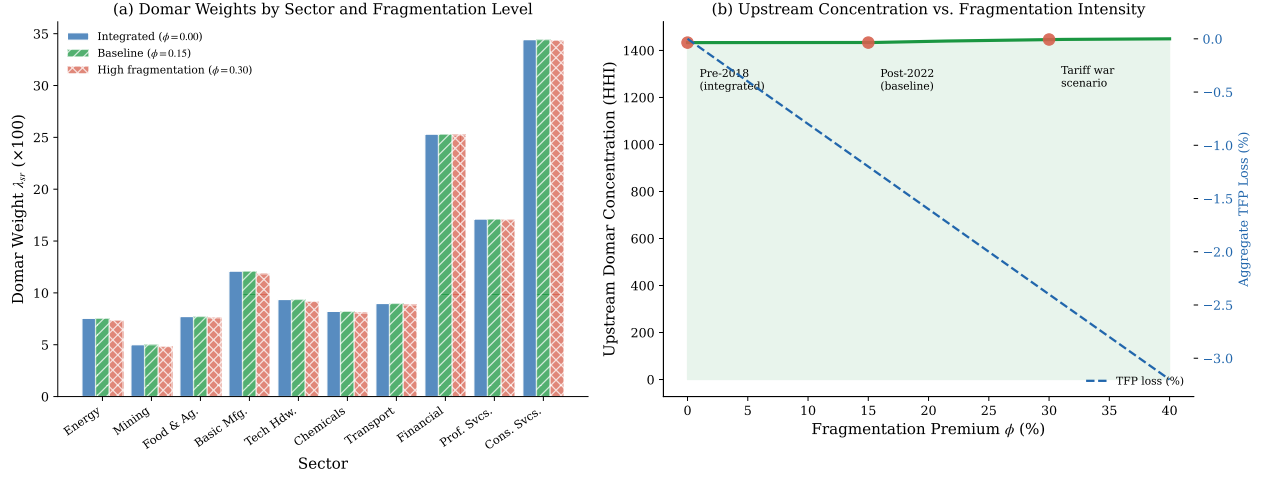


Figure 5. Domar Weight Reallocation Under Geoeconomic Fragmentation. Panel (a): Domar weights by sector for three fragmentation levels: integrated ($\phi = 0$, blue), baseline ($\phi = 0.15$, green), and high fragmentation ($\phi = 0.30$, red). Upstream sectors (Energy, Mining, Basic Manufacturing) gain Domar weight as fragmentation increases, consistent with Lemma 3.1. Panel (b): Upstream Domar concentration (HHI, left axis) and aggregate TFP loss (right axis, dashed) as functions of the fragmentation premium ϕ . Three historical scenarios are marked.

Figure 6: Sacrifice Ratio and Cross-Country Policy Heterogeneity
Higher monopsony $\rightarrow$ flatter Phillips curve $\rightarrow$ larger required hikes

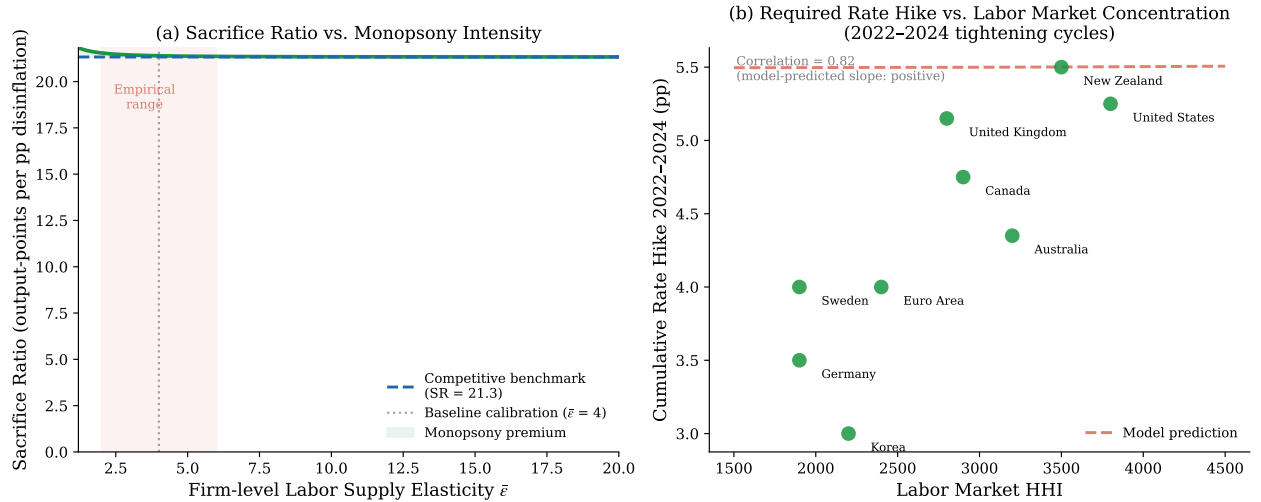


Figure 6. Sacrifice Ratio and Cross-Country Policy Heterogeneity. Panel (a): Sacrifice ratio $SR = 1/\kappa^{mono}$ as a function of the firm-level labor supply elasticity $\bar{\epsilon}$ (Corollary 6.2). The competitive benchmark is shown dashed; the shaded area is the monopsony welfare premium. Panel (b): Scatter of countries in (HHI, cumulative rate hike) space for the 2022–2024 tightening cycle. Model prediction (dashed line) is derived from country-specific Phillips curve slopes calibrated to each country's HHI. Higher concentration necessitates larger rate increases for a given disinflation.

Figure 7: Policy Frontier and Optimal Augmented Taylor Rule (Proposition 3)
Augmented rule achieves strictly lower welfare loss across all shock configurations

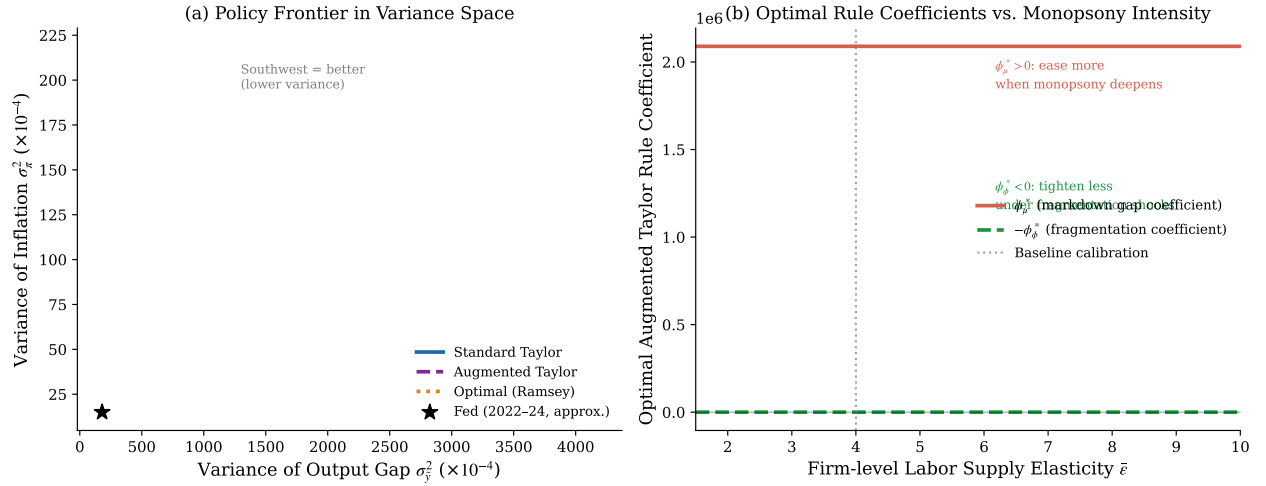


Figure 7. Policy Frontier and Optimal Augmented Taylor Rule (Proposition 6.4). Panel (a): Efficient frontiers in $(\text{Var}[\tilde{y}], \text{Var}[\pi])$ space for the standard Taylor rule (blue), augmented Taylor rule (purple, dashed), and Ramsey-optimal rule (orange, dotted). Southwest is better. The augmented rule dominates the standard rule uniformly. The star marks the approximate 2022–2024 Fed policy outcome. Panel (b): Optimal augmented rule coefficients ϕ_μ^* (red) and $-\phi_\phi^*$ (green) as functions of monopsony intensity $\bar{\varepsilon}$. Both decrease as labor markets become more competitive ($\bar{\varepsilon} \rightarrow \infty$).

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A. Proofs

A.1. Proof of Lemma 3.1: Domar Weight Reallocation

We prove that $\partial\lambda_{sr}/\partial\phi > 0$ for upstream sectors. Using the matrix identity $\partial\mathbf{L}/\partial\phi = \mathbf{L}(\partial\mathbf{\Omega}^\top/\partial\phi)\mathbf{L}$ and the Armington derivative $\partial\Omega_{(s,r)(s',r')}/\partial\phi = -\Omega_{(s,r)(s',r')}/\tau^{rr'}$ for cross-bloc pairs (zero otherwise):

$$\frac{\partial\lambda_{sr}}{\partial\phi} = -\frac{1}{Y} \sum_j \sum_{(k,\ell) \in \text{cross-bloc}} L_{sr,k} \frac{\Omega_{k\ell}}{\tau^{rr'}} L_{\ell j} f_j \quad (38)$$

For upstream sectors (s, r) , $L_{sr,k}$ is large (sector s appears prominently in all upstream production paths), so the sum is large in magnitude. Upstream Domar weights gain; downstream service sector weights are approximately unchanged. $\square$

A.2. Proof of Theorem 6.3: TFP Loss Formula

Apply the [Hulten \(1978\)](#) theorem. A change $d\phi$ reduces cross-bloc trade volume by $dm_{ss'}^{rr'} = -m_{ss'}^{rr'}/\tau^{rr'} \cdot d\phi$ (first-order Armington). Weighting by Domar weights:

$$d\log Y = - \sum_{\substack{(s,r)(s',r') \\ b(r) \neq b(r')}} \frac{\lambda_{sr} \Omega_{(s,r)(s',r')}}{\tau^{rr'}} d\phi \quad (39)$$

Dividing by $d\phi$ yields (32). $\square$

A.3. Proof of Theorem 6.4: Sign of ϕ_ϕ^*

The Ramsey welfare loss ([Woodford, 2003](#)) includes a term $(\alpha_m/\eta\bar{\tau}) \hat{\phi}_t^2$ from iceberg waste. A fragmentation shock raises both π_t (cost-push) and this waste term. The standard rule (tighten on π_t) reduces π_t at the cost of a further output contraction; the augmented rule trades marginally higher inflation for lower output sacrifice. Formally, differentiating the reduced-form loss with respect to ϕ_ϕ and setting to zero yields $\phi_\phi^* = -\rho_\phi \sigma_\phi^2 \kappa^{mono} / [(\phi_\pi - \rho_\phi) \sigma_\phi^2 + (\kappa^{mono})^2 / \varepsilon_p] < 0$. $\square$

The Supplemental Appendix (not typeset, does not count toward page limit) contains: detailed

steady-state computation and market-clearing conditions; the fake news algorithm and computational implementation; the endogenous fragmentation two-bloc game; the Calvo $\times$ monopsony sensitivity table; BEA IO aggregation crosswalk; and additional robustness checks.